



SARTHAK FULZELE

Full Stack AI-Developer, Independent Projects

Bachelor and Master of Technology in Mechanical Engineering, IIT Kharagpur

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EDUCATION

Bachelor & Master of Technology, IIT Kharagpur

2021 - 2026

XII (Science), Vidya Vihar Convent High School, Chandrapur

2019 - 2021

X, BJM Carmel Academy, Chandrapur

2009 - 2019

WORK EXPERIENCE

[AI-Engineer Intern](#) | [Curve-AI](#)

Jan 2025 – Apr 2025

Python, Flask, LangGraph, Docker, AWS EC2, AssemblyAI, GPT-4o-mini, ElevenLabs, Google Calendar API, AsyncIO, Multiprocessing

Objective: Developed a production-ready voice AI assistant focused on natural conversations, automation, and scalable deployment.

- Built a real-time voice chatbot by integrating **AssemblyAI**, **GPT-4o-mini**, and **ElevenLabs** for natural, human-like dialogue
- Developed and containerized **Flask** backend APIs, deployed them on **AWS EC2**, and optimized **async I/O** with **multiprocessing**
- Implemented interruption handling and Google Calendar automation to support fluid turn-taking and seamless scheduling
- Collaborated on system architecture improvements to enhance reliability, scalability, and **low-latency** system performance

[Research Internship](#) | [Deakin University](#) | Prof. Ye Zhu and Prof. Ming Liu

Aug 2025 - Dec 2025

Python, PyTorch, LLaVA, LoRA, OpenCV, Pandas, NumPy, NLTK, Streamlit, Medical VQA, Model Evaluation Pipelines

Objective: Researched medical AI systems for diagnosis and support through fine-tuning, evaluation, and deployment pipelines.

- Researched multimodal LLMs for disease diagnosis, decision support, & scalable real-world clinical deployment use cases today
- Preprocessed 150k MedTrinity image-text pairs with **Python**, **OpenCV**, **Pandas**, **NumPy**, and **NLTK** for **LLaVA fine-tuning** pipelines
- Implemented **LoRA** and projector tuning on **LLaVA**, improving accuracy and F1-score on medical VQA benchmarks
- Built evaluation and **Streamlit** chatbot pipelines with clinician feedback, automated metrics, and interpretable clinical outputs

SKILLS

Languages:	• Golang	• Python	• C++	• Typescript	• Javascript	• SQL	
Frameworks:	• React.js	• Express.js	• Next.js	• Flask	• Django	• FastAPI	• Socket.io
Databases:	• PostgreSQL	• MongoDB	• MySQL	• Redis	• Elasticsearch		
Cloud & DevOps:	• Docker	• Kubernetes	• Kafka	• AWS	• Git		

PROJECTS

[JAMIFY](#) | Full-Stack Development <https://jamify-nly7.onrender.com>

ReactJs, Redux, Tailwind, NodeJs, ExpressJs, MongoDB, REST API, SocketIO

Objective: Built a real-time chat and collaborative music platform with synchronized playback and shared jam rooms for multiple users.

- Built a **real-time chat** and collaborative music platform enabling shared jam rooms and synchronized playback for multiple users
- Built secure **REST APIs** for authentication, messaging, playlist management, liked songs, and live multi-user session workflows
- Implemented **JWT-based authentication** and role-based access control to secure user accounts and broader the platform flows
- Built a responsive **React** SPA for chat, music discovery, playlists, and jam rooms with persistent global state in **Redux** Toolkit
- Implemented **Socket.IO** features for messaging, presence, typing indicators, read receipts, and synchronized playback for rooms
- Designed **synchronized controls** for play, pause, seek, track switching, and queue updates across shared **multi-user jam** sessions

[TWO-WAY REAL-TIME DATA INTEGRATION PLATFORM](#) | Event-Driven Systems & System Integration

FastAPI, Apache Kafka, MySQL, Stripe API & Webhooks, Docker, Python

Objective: Engineered an event-driven synchronization platform that maintained real-time consistency between internal customer data and Stripe using Kafka, webhooks, and asynchronous processing.

- Built a **FastAPI** service to stream customer CRUD events from MySQL to Stripe via **Apache Kafka**, sustaining **1,000+ events/min**
- Created Stripe **webhook** ingestion with HMAC signature verification & idempotent event handling, eliminating duplicate writes
- Decoupled HTTP APIs from external integrations using Kafka consumers, reducing API response **latency to <50 ms**
- Containerized FastAPI, MySQL, Kafka, and Zookeeper with **Docker Compose**, creating reproducible local environments in <2 mins

[BLOGGING SERVICE](#) | Distributed Systems & Backend Engineering

Go, Kafka, Elasticsearch, Docker, Kubernetes, REST APIs

Objective: Built a scalable microservices blogging platform enabling asynchronous content publishing, real-time indexing, & fast search

- Built a **Go microservices** blogging platform with **Kafka & Elasticsearch**, supporting **concurrent** publishing and low-latency search
- Designed an async Kafka ingestion pipeline to decouple blog writes from indexing, to improve API responsiveness under load
- Built a Kafka consumer service that processed blog events & indexed documents into Elasticsearch, keeping it stable under load
- Implemented a search API with keyword filters and relevance-ranked retrieval across blog posts for low-latency content discovery
- Containerized services with **Docker** & deployed them on **Kubernetes** with **scalable** pods, service discovery, and safe **rollout flows**
- Collaborated on service contracts, error handling, and observability to keep the platform stable for production-grade operations